

Walmart Sales Analysis

Business Problem Statement

Walmart generates thousands of daily transactions across multiple branches, cities, product categories, and payment methods. Without proper analysis, it becomes difficult for management to identify customer purchasing behavior, profitable product categories, branch performance, payment preferences, customer satisfaction, and revenue trends. This project aims to transform raw sales data into meaningful business insights using an ELT pipeline, enabling data-driven decisions to improve sales performance, operational efficiency, and profitability.

Project Overview

This Walmart Sales Analytics project demonstrates an ELT (Extract, Load, Transform) workflow using Python (Pandas) and MySQL. Raw sales data is extracted and cleaned using Python, loaded into MySQL, and transformed through SQL queries to generate business insights. The analysis focuses on customer behavior, branch performance, sales trends, profitability, ratings, and payment preferences. The project showcases practical data engineering and SQL analytics skills suitable for a data analyst portfolio.

Project Deliverables

- Business Problem Statement
- Project Overview
- Raw Walmart Sales Dataset (CSV)
- Python Data Extraction & Cleaning Notebook
- MySQL Database & Table Creation Scripts
- Data Loading Scripts
- SQL Transformation Queries
- Business Analysis SQL Queries
- Key Business Insights
- Business Recommendations

Data Collection

Collect raw Walmart sales transaction data from an API

```
import pandas as pd
from kaggle.api.kaggle_api_extended import KaggleApi

api = KaggleApi()
api.authenticate()

print("Kaggle API connected successfully")
```

✓ 8.3s

Kaggle API connected successfully

Data Extraction (Python)

Import the CSV file into Python using Pandas for initial exploration and validation.

```
api.dataset_download_files(
    'najir0123/walmart-10k-sales-datasets',
    path='./data',
    unzip=True
)

print("Successfully imported")
```

✓ 5.0s

Dataset URL: <https://www.kaggle.com/datasets/najir0123/walmart-10k-sales-datasets>
Successfully imported

Loading datasets using pandas

```
df = pd.read_csv('data/Walmart.csv')
print('Datasets loaded successfully')
```

✓ 0.3s

Datasets loaded successfully

Data Profiling

Analyze the dataset structure

```
df.head()
```

✓ 1.4s

	invoice_id	branch	city	category	unit_price	quantity	date	time	payment_method	rating	profit_margin	total_sale
0	1	WALM003	San Antonio	Health and beauty	74.69	7.0	2019-01-05	13:08:00	Ewallet	9.1	0.48	522.83
1	2	WALM048	Harlingen	Electronic accessories	15.28	5.0	2019-03-08	10:29:00	Cash	9.6	0.48	76.40
2	3	WALM067	Haltom City	Home and lifestyle	46.33	7.0	2019-03-03	13:23:00	Credit card	7.4	0.33	324.31
3	4	WALM064	Bedford	Health and beauty	58.22	8.0	2019-01-27	20:33:00	Ewallet	8.4	0.33	465.76
4	5	WALM013	Irving	Sports and travel	86.31	7.0	2019-02-08	10:37:00	Ewallet	5.3	0.48	604.17

Analyze data types

```
df.info()
✓ 2.2s

<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   invoice_id            10000 non-null  int64
1   branch                10000 non-null  str
2   city                  10000 non-null  str
3   category              10000 non-null  str
4   unit_price            9969 non-null   float64
5   quantity              9969 non-null   float64
6   date                  10000 non-null  datetime64[us]
7   time                  10000 non-null  object
8   payment_method        10000 non-null  str
9   rating                10000 non-null  float64
10  profit_margin         10000 non-null  float64
11  total_sale            9969 non-null   float64
dtypes: datetime64[us](1), float64(5), int64(1), object(1), str(4)
memory usage: 937.6+ KB
```

Checking missing values

```
df.isnull().sum()

invoice_id      0
Branch          0
City            0
category        0
unit_price      31
quantity        31
date            0
time            0
payment_method  0
rating          0
profit_margin   0
dtype: int64
```

Checking duplicate records

```
df.duplicated().sum()
✓ 0.1s

np.int64(51)
```

Data Cleaning

Dropping Duplicates rows

```
df.drop_duplicates(inplace=True)
```

✓ 0.0s

Maintain column names in lower case

```
df.columns = df.columns.str.lower()
```

✓ 0.0s

Removing currency symbol & convert data types

```
df['unit_price'] = df['unit_price'].str.replace('$', '').astype(float)
```

✓ 0.0s

Adding Calculated column

```
df['total_sale'] = df['unit_price'] * df['quantity']
```

df

✓ 0.0s

Changing date and time format

```
df['date'] = pd.to_datetime(df['date'], format='%d-%m-%Y')  
# convert time column  
df['time'] = pd.to_datetime(df['time'], format='%H:%M:%S').dt.time
```

✓ 0.6s

Database Creation

Create a Schema in MySQL database to store the cleaned Walmart sales data.

Table Design

Design the Walmart sales table with appropriate columns, data types, and primary key constraints.

```
CREATE TABLE sales.walmart_data (  
    invoice_id int primary key,  
    branch varchar(50),  
    city varchar(50),  
    category varchar(50),  
    unit_price decimal(10,2),  
    quantity decimal(10,2),  
    date date,  
    time Time,  
    payment_method varchar(50),  
    rating decimal(10,2),  
    profit_margin decimal(10,2),  
    total_sale decimal(10,2)  
) ;
```

Data Loading

Load the cleaned dataset from Python into the MySQL database.

```
import pymysql  
from sqlalchemy import create_engine  
  
engine = create_engine(  
    "mysql+pymysql://root:spatil123@localhost:3306/sales"  
)  
  
print("Connected Successfully")  
✓ 3.3s
```

Connected Successfully

```
df.to_sql(  
    name='walmart_data',  
    con=engine,  
    if_exists='append',  
    index=False  
)  
  
print("Data Loaded Successfully")
```

Data Validation

Verify that all records have been successfully loaded without data loss or duplication.

```
SELECT * FROM sales.walmart_data LIMIT 0, 50000
```

10000 row(s) returned

Result Grid						
	Field	Type	Null	Key	Default	Extra
►	invoice_id	int	NO	PRI	NULL	
	branch	varchar(50)	YES		NULL	
	city	varchar(50)	YES		NULL	
	category	varchar(50)	YES		NULL	
	unit_price	decimal(10,2)	YES		NULL	
	quantity	decimal(10,2)	YES		NULL	
	date	date	YES		NULL	
	time	time	YES		NULL	
	payment_method	varchar(50)	YES		NULL	
	rating	decimal(10,2)	YES		NULL	
	profit_margin	decimal(10,2)	YES		NULL	
	total_sale	decimal(10,2)	YES		NULL	

Data Transformation (SQL)

Final Data for analysis

Result Grid												
	invoice_id	branch	city	category	unit_price	quantity	date	time	payment_method	rating	profit_margin	total_sale
►	1	WALM003	San Antonio	Health and beauty	74.69	7.00	2019-01-05	13:08:00	Ewallet	9.10	0.48	522.83
	2	WALM048	Harlingen	Electronic accessories	15.28	5.00	2019-03-08	10:29:00	Cash	9.60	0.48	76.40
	3	WALM067	Haltom City	Home and lifestyle	46.33	7.00	2019-03-03	13:23:00	Credit card	7.40	0.33	324.31
	4	WALM064	Bedford	Health and beauty	58.22	8.00	2019-01-27	20:33:00	Ewallet	8.40	0.33	465.76
	5	WALM013	Irving	Sports and travel	86.31	7.00	2019-02-08	10:37:00	Ewallet	5.30	0.48	604.17
	6	WALM026	Denton	Electronic accessories	85.39	85.39	2019-03-25	18:30:00	Ewallet	4.10	0.48	597.73
	7	WALM088	Cleburne	Electronic accessories	68.84	6.00	2019-02-25	14:36:00	Ewallet	5.80	0.33	413.04
	8	WALM100	Canyon	Home and lifestyle	73.56	10.00	2019-02-24	11:38:00	Ewallet	8.00	0.18	735.60
	9	WALM066	Grapevine	Health and beauty	36.26	2.00	2019-01-10	17:15:00	Credit card	7.20	0.33	72.52
	10	WALM065	Texas City	Food and beverages	54.84	3.00	2019-02-20	13:27:00	Credit card	5.90	0.33	164.52
	11	WALM013	Irving	Fashion accessories	14.48	4.00	2019-02-06	18:07:00	Ewallet	4.50	0.48	57.92
	12	WALM035	San Angelo	Electronic accessories	25.51	4.00	2019-03-09	17:03:00	Cash	6.80	0.48	102.04
	13	WALM027	Abilene	Electronic accessories	46.95	5.00	2019-02-12	10:25:00	Ewallet	7.10	0.48	234.75
	14	WALM035	San Angelo	Food and beverages	43.19	10.00	2019-02-07	16:48:00	Ewallet	8.20	0.48	431.90
	15	WALM031	Lewisville	Health and beauty	71.38	10.00	2019-03-29	19:21:00	Cash	5.70	0.48	713.80
	16	WALM008	Corpus Christi	Sports and travel	93.72	6.00	2019-01-15	16:19:00	Cash	4.50	0.48	562.32
	17	WALM029	Round Rock	Health and beauty	68.93	7.00	2019-03-11	11:03:00	Credit card	4.60	0.48	482.51
	18	WALM061	Cedar Park	Sports and travel	72.61	6.00	2019-01-01	10:39:00	Credit card	6.90	0.33	435.66
	19	WALM053	Conroe	Food and beverages	54.67	3.00	2019-01-21	18:00:00	Credit card	8.60	0.57	164.01
	20	WALM013	Irving	Home and lifestyle	40.30	2.00	2019-03-11	15:30:00	Ewallet	4.40	0.48	80.60
	21	WALM044	Pharr	Electronic accessories	86.04	5.00	2019-02-25	11:24:00	Ewallet	4.80	0.48	430.20
	22	WALM088	Cleburne	Health and beauty	87.98	3.00	2019-03-05	10:40:00	Ewallet	5.10	0.33	263.94

Business Analysis

1. Identify different payment methods and their transaction counts & No of quantity sold

Result Grid			
	payment_method	Num_of_trans	total_qty
►	Ewallet	3911	8932.00
	Cash	1832	4984.00
	Credit card	4257	9567.00

- Find the Average highest-rated product category in each branch.

Result Grid   Filter Rows: Export:   Wri

	Branch	category	Average_rating	rnk
▶	WALM001	Electronic accessories	7.450000	1
	WALM002	Food and beverages	8.250000	1
	WALM003	Sports and travel	7.500000	1
	WALM004	Food and beverages	9.300000	1
	WALM005	Health and beauty	8.366667	1
	WALM006	Fashion accessories	6.797059	1
	WALM007	Food and beverages	7.550000	1
	WALM008	Food and beverages	7.400000	1
	WALM009	Sports and travel	9.600000	1
	WALM010	Electronic accessories	9.000000	1
	WALM011	Food and beverages	7.000000	1
	WALM012	Health and beauty	7.450000	1
	WALM013	Health and beauty	7.600000	1
	WALM014	Electronic accessories	6.833333	1

- Determine the busiest day of the week for every branch based on transaction volume?

	branch	Busiest_day	trans
▶	WALM001	Thursday	16
	WALM002	Thursday	15
	WALM003	Tuesday	33
	WALM004	Sunday	14
	WALM005	Wednesday	19
	WALM006	Thursday	15
	WALM007	Friday	12
	WALM007	Sunday	12
	WALM008	Tuesday	17
	WALM009	Sunday	42
	WALM010	Wednesday	12
	WALM010	Friday	12
	WALM011	Tuesday	18
	WALM012	Sunday	20
	WALM013	Monday	10

- Analyze quantity sold by payment method across categories.

Result Grid			
	Filter Rows:		Export:
	payment_method	category	total_qty
▶	Cash	Electronic accessories	449.00
	Cash	Fashion accessories	1806.00
	Cash	Food and beverages	316.00
	Cash	Health and beauty	283.00
	Cash	Home and lifestyle	1794.00
	Cash	Sports and travel	336.00
	Credit card	Electronic accessories	689.00
	Credit card	Fashion accessories	3973.00
	Credit card	Food and beverages	320.00
	Credit card	Health and beauty	288.00
	Credit card	Home and lifestyle	3998.00
	Credit card	Sports and travel	299.00
	Ewallet	Electronic accessories	356.00
	Ewallet	Fashion accessories	3874.00
	Ewallet	Food and beverages	316.00

- Calculate average, minimum, and maximum customer ratings by city?

	City	Average_rating	Min_rating	Max_rating
▶	San Antonio	5.370430	3.00	9.90
	Harlingen	6.512500	3.00	9.60
	Haltom City	6.404054	3.00	9.90
	Bedford	6.331250	3.00	9.30
	Irving	6.291228	3.00	9.80
	Denton	6.678082	3.00	9.20
	Cleburne	6.468354	3.00	9.60
	Canyon	6.208065	3.00	9.00
	Grapevine	6.294667	3.00	9.00
	Texas City	5.042045	3.00	8.60
	San Angelo	5.227746	3.00	9.30
	Abilene	6.336111	4.00	9.70
	Lewisville	6.244643	3.00	9.30
	Corpus Christi	6.388235	3.00	9.70
	Round Rock	5.238265	3.00	9.90
	Cedar Park	6.569565	3.00	9.90
	Conroe	6.398649	3.00	9.20
	Pharr	6.198571	3.00	9.90
	Farmers Br...	6.215278	3.00	9.00
	Midland	6.421429	3.00	9.70
	Houston	6.336000	3.00	9.50
	Lancaster	6.057143	3.00	9.80

- Rank product categories based on total profit generated.

	category	total_profit
▶	Fashion accessories	192314.8932
	Home and lifestyle	192213.6381
	Electronic accessories	30772.4895
	Food and beverages	21552.8622
	Sports and travel	20613.8082
	Health and beauty	18671.7345

7. Identify the most frequently used payment method in each branch.

	branch	payment_method	total_count
▶	WALM001	Ewallet	46
	WALM001	WALM001	29
	WALM002	Ewallet	37
	WALM002	Credit card	26
	WALM002	Cash	2
	WALM003	Credit card	115
	WALM003	Cash	56
	WALM003	Ewallet	15
	WALM004	Ewallet	44
	WALM004	Credit card	14
	WALM004	Cash	2
	WALM005	Ewallet	56
	WALM005	Credit card	19
	WALM005	Cash	9
	WALM006	Ewallet	50
	WALM006	Credit card	19
	WALM006	Cash	2
	WALM007	Ewallet	52
	WALM007	Credit card	13
	WALM007	Cash	5
	WALM008	Ewallet	39
	WALM008	Credit card	25

8. Analyze transaction distribution across Morning, Afternoon, and Evening shifts.

	shifts	total_trans
▶	Afternoon	4662
	Morning	2095
	Evening	3243

9. Find top 5 branches experienced the largest decrease in revenue compared to the previous year?(Current year 2023 to previous year 2022)

	branch	Sale_2023	Sale_2022	revenue_decreases_in_%
▶	WALM045	647.00	1731.00	62.62
	WALM047	1069.00	2581.00	58.58
	WALM098	1030.00	2446.00	57.89
	WALM033	931.00	2099.00	55.65
	WALM081	850.00	1723.00	50.67

Insight Generation

- Credit Card was the most preferred payment method, recording 4,257 transactions and the highest quantity sold of 9,567 units.
- Health and Beauty was the highest-rated product category, indicating strong customer satisfaction.
- WALM009 recorded the highest branch activity with 42 transactions on Sunday, making it the busiest branch day combination.
- Home and Lifestyle was the top selling category across all payment methods, reflecting consistent customer demand.
- Fashion Accessories generated the highest profit of 192,314, making it the most profitable product category.

- The Afternoon (12:00 PM–6:00 PM) shift recorded the maximum number of transactions, representing the peak shopping period.
- WALM045 experienced the highest year-over-year revenue decline, indicating a need for focused performance improvement.

Business Recommendations

- Promote credit card offers to increase sales, as it is the most preferred payment method.
- Expand the Health and Beauty product range to capitalize on its high customer satisfaction.
- Increase staff and inventory on Sundays, especially at high performing branches like WALM009, to handle peak demand.
- Maintain higher stock levels for Home and Lifestyle products to avoid stockouts during periods of high demand.
- Focus marketing efforts on Fashion Accessories to maximize profits from the highest-performing category.
- Schedule more employees during the afternoon shift (12 PM–6 PM) to improve customer service during peak shopping hours.
- Investigate the decline in revenue at WALM045 and implement targeted promotions, pricing strategies, or operational improvements to restore sales.